7 August 1959

MILITARY SPECIFICATION

RANGEFINDER MARK 75 MOD 1

1. SCOPE

1.1 Scope. - This specification covers the Rangefinder Mark 75 Mod 1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-M-13508 - Mirrors, Front Surfaced, Aluminum, for Optical Elements.
MIL-G-15793 - Grease, Instrument.
MIL-G-16908 - Grease, Bearing (for General Ordnance Use).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

BUREAU OF ORDNANCE

LD 412545 - Rangefinder Mark 75 Mod 1, Assembled Units; List of Drawings, Assemblies, Parts, Specifications, Etc.
LD 413777 - Stowage Box, For Rangefinder Mark 75; List of Drawings, Assemblies, Parts, Specifications, Etc.

FSC 1240

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PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.
NAVORD OD 13521 - Rangefinder Mark 75 Mod 1, Factory Acceptance Test.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. - Requirements for manufacture shall be in accordance with Publication OP 400. The rangefinder shall conform to LD 412545 and all documents listed thereon, this specification, and the contract or order.

3.2 Preproduction sample. - Unless otherwise specified in the contract or order (see 6.2), a preproduction sample is required (see 4.3.1.1) for each such contract or order. Samples shall be manufactured using the methods, procedures, and plant location proposed for the production lot. Samples shall be inspected as specified in 4.5.1, such inspection being for the purpose of determining that the manufacturer's production process will produce rangefinders that will meet the requirements of this specification.

3.3 Construction. -

3.3.1 Cleanness. - The entire interior of the gas-tight compartment and of the eyepiece assemblies shall be free from oil, dust, dirt, and foreign matter of any kind, and free from any lubricant that might in time collect on the surface of any optical part. Optical components shall be carefully cleaned before being assembled into the instrument to insure freedom from dust, dirt, moisture, fingerprints, and other contamination. Reagents used for cleaning cemented optical components shall not be of a type that would be absorbed into the cement and cause subsequent deterioration of the cement or cause cement starts. Scales shall be clean and legible.

3.3.2 Optical cementing. - The cement film binding optical elements together shall be free from foreign particles, bubbles, cement starts, milkiness, and other defects that might reduce the effectiveness of the rangefinder. Excess cement shall be removed from the exterior of cemented parts; however, no reagent shall be used that could be absorbed into the cement film or cause cement starts. All cemented lenses shall be marked on the edge with the symbol "CC" indicating cellulose caprate cement.

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3.3.3 End reflector surfaces. - Aluminizing and overcoating of end reflectors shall be in accordance with Specification MIL-M-13508 except that silicon monoxide overcoating shall be used in lieu of magnesium fluoride.

3.3.4 Lubrication. - All gearing, sliding mechanisms, bearings, and moving joints shall be lubricated with instrument grease conforming to Specification MIL-G-15793. Care shall be taken that the optical parts remain free of grease and any vapor therefrom.

3.3.5 Reticles. - When seen through the eyepieces, the markings of the reticles shall be sharp and clean-cut. There shall be no fillets at the intersections of lines or other imperfections visible. No lines shall appear indistinct or broken on account of improper filling. There shall be no perceptible difference in the size, form, or appearance of any of the stereoscopic reference marks in one field and the corresponding marks in the other field.

3.3.6 Main outer bearings. - The main outer bearings shall receive their final machining after assembly on the outer tube in order to meet the bearing concentricity requirements of the applicable drawings.

3.3.7 Stray Light. - No stray light shall be visible in or surrounding the fields of view, nor shall there be false pupils or other concentrations of stray light in or adjacent to the exit pupils. Except as specifically required, no paint, enamel, lacquer, or other material that might become detached from the surface to which applied shall be used in the interior of these rangefinders.

3.3.8 Artificial illumination. - The illumination of the stereoscopic reference marks and of the internal-adjuster target images in each field shall be such that all of the marks are distinctly visible under all conditions when illumination is needed, all of the stereoscopic reference marks are of substantially the same brightness, the adjuster target images are of the same brightness, and the remainder of the field is dark with no bright or otherwise troublesome reflection surrounding the field. The brightness of corresponding marks, including the adjuster target images, in the two fields shall be substantially the same at any setting of the controlling rheostat. The illumination of the scales shall show clearly the index and that part of the scale adjacent thereto including at least two graduation numbers on either side of the index. There shall be no bright or otherwise troublesome reflection and no bright or otherwise troublesome reflection and no bright light or reflection shall be visible at a distance in the dark.

3.3.9 Rotational balance. - The rangefinder, with elevating gear attached, shall be in static balance about its longitudinal axis in all positions of rotation of the rangefinder.

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3.3.10 Gastightness. - Under conditions of approximate constant temperature and barometric pressure, the internal gas pressure drop in the rangefinder shall not exceed 1 pound per square inch (psi) from an initial internal pressure of 4 psi during a period of not less than 8 hours. When the internal gas pressure is increased or decreased the rangefinder shall remain internally clean and mechanical operation and optical performance shall remain unchanged.

3.3.11 Charging. - The rangefinder shall be charged with helium of 98 percent purity to a pressure of 4 psi.

3.4 Mechanical requirements. -

3.4.1 Operation of mechanisms. - All mechanisms involved in optical settings and readings shall operate smoothly throughout their entire range of operation. Detenting mechanisms shall detent in required positions without play.

3.4.2 Stability of settings. - Settings made on the range mechanisms and correction wedge shall be stable when the rangefinder is elevated and depressed throughout its entire range of elevation. The height adjustment for any one range shall be correct for any other range when viewed immediately after the adjustments have been made.

3.4.3 Change of range mechanism. - The change of range mechanism shall be accurate to the extent that the error in yards between the transmitter and the inner range scale reading shall not exceed the value calculated from:

$$E = \frac{R}{300} + 9.1$$

where E = maximum allowable error in yards
R = range in yards

3.4.4 Range knob torque. - When the rangefinder is stabilized at an ambient temperature of approximately 70°F, the torque required to rotate the range knob and drive the compensator wedges shall not exceed 20 inch-ounces in either direction of rotation of the range knob at any range reading.

3.4.5 Orientation of filters. - The filters shall be assembled so that when the filter mount is actuated the same kind of filter (or clear glass) is positioned in the two optical systems. The variable density filters shall be set so that counter-clockwise rotation of filter knob increases the density and the positions of maximum and minimum density are reached just before the knob reaches its limits of travel.

3.5 Alignment and optical performance. -

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3.5.1 Image quality. - The image of a round bright star shall appear round with a sharply defined edge under the test conditions of 4.5.4.1. The image shall be free from points, flares, rays, fog, haze, or halos other than the normal diffraction rings. No difference in coloration between one side of the star and the other shall be seen. A slightly defocused image shall remain round. The ocular setting between the three stars shall not exceed 0.2 diopter. These requirements shall be met in each telescope and at each magnification.

3.5.2 Resolution. - The limit of resolution shall not exceed two seconds of arc in both horizontal and vertical directions simultaneously, under the test conditions of 4.5.4.1. This requirement shall be met in each telescope and at each magnification.

3.5.3 Alignment of line of sight. - The optical systems shall be so aligned in the rangefinder tube that the left line of sight of the rangefinder, as determined by the projection of the measuring mark in the left telescopic system, shall be perpendicular to the axis of the bearing surfaces between the flanges of the bearing rings within two minutes of arc.

3.5.4 Parallelism of emergent rays. - At any setting of the interpupillary scale and of either of the diopter scales, and at either magnification, the axis of the pencil of light from the measuring mark in either telescopic system, after emergence from the ocular of that system, shall be parallel to the axis of the corresponding pencil emerging from the other ocular, within the following limits:

(a) Ten minutes arc in a plane perpendicular to the axis of the rangefinder. This is equivalent to 25 seconds at the higher magnification and to 50 seconds at the lower magnification.

(b) These axes, as they proceed from the oculars, shall not converge, and their divergence shall not exceed 30 minutes. This is equivalent to one minute fifteen seconds at the higher magnification and to two minutes thirty seconds at the lower magnification.

3.5.5 Tilt of fields. - The images of a distant straight line formed by the two optical systems shall be parallel within 0.25 degree. Also the difference in apparent vertical position of corresponding marks in the two fields of view shall be equal to the apparent difference in vertical positions of the measuring marks, as defined in 3.5.4, and in the same direction, within the following limits:

(a) Two minutes of apparent field for the two marks at five mils and the 0, 50, and 100 unit of error marks.

(b) Four minutes apparent field for the two marks at ten mils.

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3.5.6 Equality of magnification. - The magnifications of the right and left-handed telescopes shall be equal within 1 percent.

3.5.7 Coincidence of focal planes of outside object and reticle markings (freedom from parallax). - The best focal setting for an outside target at the center of the field, the measuring mark, and the internal-adjuster target shall not differ from that of the image of any other of these in the same field by more than 0.2 diopter. The limit shall not be exceeded when the focus of any of these images, at either magnification, is compared with the focus of any other image at the same or the other magnification. The requirements shall be met for both right and left hand fields, and the residual parallax between the two fields shall be such that when the rangefinder is directed at a target at least 3000 yards away, so that the target image is near any stereoscopic reference mark on the horizontal diameter of the field, vertical or horizontal motion of the range-taker's eye shall cause no perceptible change in the apparent relative distance of target and mark as seen stereoscopical

3.5.8 Interpupillary adjustment. - Each graduation of the scale shall give the correct interpupillary distance within 1/2 millimeter. The interpupillary distance shall not be affected by changes in the focusing motion.

3.5.9 Accuracy of diopter graduation. - Each diopter mark on the main and range scale eyepiece shall be accurately located with respect to sharp focus on the measuring mark or scale index within 1/4 diopter. For the main eyepiece this requirement shall be met in both high and low power and with any of the color or polarizing filters or the clear filter in operation.

3.5.10 Correction scale eyepiece focus. - The correction scale eyepiece shall be focused sharply on the correction scale index.

3.5.11 Alignment of inner and outer range scales. - Inner and outer range scales shall be so aligned that the difference in readings between the two scales for any range value, when the scales are moved in either increasing range or decreasing range, shall not exceed one-half unit of error (see 6.3) at high power.

3.5.12 Reproducibility of internal-adjuster settings. - The quality of the internal adjuster system shall be such that the spread of a set of 5 readings made in the direction of decreasing adjuster scale readings shall not exceed 1 unit of error. Also, the mean of 10 readings obtained by making the final setting in one direction shall not differ by more than 1/2 unit of error from the mean of 10 readings made by making the final setting in the other direction.

3.5.13 Reproducibility of range settings. - The optical quality of the rangefinder shall be such that the mean variation from the mean of ten readings taken at the measuring mark on targets at known ranges of not less than 3000 yards shall not exceed one unit of error.

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3.5.14 Ranging accuracy. - The mean of ten single range readings shall not vary from the true range by more than the limiting values given in Table I.

Table I. - Maximum Allowable Errors in Mean Range Values

Magnification	True range (yards)	Error (units of error)	Error (yards)
24X	Greater than 4000	1	-
24X	Less than 4000	-	8
12X	Greater than 4000	1	-
12X	Less than 4000	-	16

3.5.15 Stability of ranging using various reticle marks. - At 24 power and when the image of the target is near the measuring mark, the mean of ten readings on a target at a range of not less than 3,000 yards shall not differ by more than 1.5 units of error from the mean of ten readings when the image is near the second or fourth stereoscopic reference mark and at the same height as the measuring mark. The mean of ten readings when the image is near the first or fifth mark shall not differ by more than 3 units of error from the mean of ten readings when near the measuring mark nor by more than 2 units of error from the mean of ten readings when near the adjacent mark. The above requirements shall be applicable to the rangefinder at 12 power except that the allowable error shall be two thirds of that specified for 24 power.

3.5.16 Dielectric strength. - The rangefinder electrical equipment and circuits shall withstand the dielectric test voltages when tested for dielectric strength in accordance with 4.5.4.1.

3.5.17 Insulation resistance. - The insulation resistance of all rangefinder electrical circuits shall be not less than 5 megohms when tested in accordance with 4.5.4.1.

3.6 Environmental requirements. -

3.6.1 Shock stability. - The rangefinder shall be capable of withstanding six 150 foot-pound blows.

3.6.2 Vibration stability. - The rangefinder shall be capable of withstanding Type I.- environmental vibration as defined in Standard MIL-STD-167 at frequencies of 5 to 15 cycles per second at an amplitude of plus or minus 0.030 inch.

3.6.3 Watertightness. - The rangefinder shall be watertight when tested in accordance with 4.5.3.3.

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3.6.4 Temperature stability. -

3.6.4.1 Temperature extremes. - All rangefinder mechanisms shall operate satisfactorily when the temperature of the rangefinder is at minus 20°F and at plus 120°F.

3.6.4.2 Temperature variation. - The mean of any set of ten readings on a given target taken at various temperatures in a range of 30°F on either side of a stabilizing temperature as stabilization is approached shall not differ from the mean of ten readings taken at the stabilized temperature, or from the mean of any other set of ten readings in a group of readings, by more than four units of error multiplied by the actual degrees F difference in temperature at the time of the readings, divided by thirty.

3.6.5 Environmental conditions. - Under environmental conditions, rangefinder construction, mechanical requirements, and alignment and optical performance shall be as specified herein.

3.7 Workmanship. - Workmanship shall be of such quality as to produce rangefinders that will meet all the requirements of the applicable drawings, standards, and specifications.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Lot. -

4.2.1 Rangefinders. - A lot shall consist of all the rangefinders manufactured by the same process and at the same plant location and submitted for Government acceptance at any one time.

4.2.2 Parts. - A lot shall consist of all like parts including spare parts manufactured by the same process and submitted for Government acceptance at any one time.

4.3 Sampling. -

4.3.1 Rangefinders. -

4.3.1.1 For preproduction inspection. - Unless otherwise specified (see 4.3.1.1.1), one preproduction sample rangefinder shall be supplied by the contractor as specified in 3.2. The sample shall be submitted to the Government inspector for preproduction inspection (see 4.5.1) at an activity designated in the contract or order (see 6.2) or by the Bureau of Ordnance. Further

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production of rangefinders by the contractor prior to the approval of the procuring activity or completion of inspection of the preproduction sample shall be at the contractor's risk. The shipping container shall be marked in accordance with 5.5.

4.3.1.1.1 Preproduction sample and inspection for a subsequent contract. - If a contractor has previously delivered rangefinders in accordance with the requirements of this specification and his product has been found to be satisfactory, the preproduction sample and inspection for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.2).

4.3.1.2 Sampling for periodic tests. - Unless otherwise specified by the procuring activity, sampling frequency for periodic tests (see 4.5.2) shall be in accordance with Table II, using the normal sample size tabulated therein except as noted in 4.3.1.2.1.

Table II. - Sampling Frequency for Periodic Tests

No. of units submitted	Sample of Size	
	Normal	Tightened
First 25 or less	1	2
Next 50 or less	1	3
Next 75 or less	1	3
Each succeeding 100 or less	1	4

The Government inspector shall forward the sample to a testing activity designated in the contract or order (see 6.2) or by the Bureau of Ordnance. Shipping containers of the sample rangefinders shall be marked in accordance with 4.3.1.1 except that containers shall be marked "Sample for periodic tests" instead of "Sample for preproduction inspection". In addition, containers shall be marked with the identification of the lot or quantity from which the sample was taken.

4.3.1.2.1 Tightened sampling. - In the event any sample fails to meet all periodic test requirements during normal sampling, tightened sampling in accordance with Table II shall be instituted and maintained until three consecutive groups of units pass the periodic tests, after which normal sampling shall be resumed.

4.3.1.3 Sampling for lot acceptance. - No provisions are made for sampling for lot acceptance since each rangefinder in each lot shall be subjected to the lot acceptance tests of 4.5.4.

4.3.2 Parts. - Sampling for lot acceptance of parts, including spare parts, shall be in accordance with the procedures of Standard MIL-STD-105.

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4.3.3 Parts containers. - A random sample of spare parts shipping containers for the inspection of 4.4.4.2 shall be selected by the Government inspector in accordance with the provisions of Standard MIL-STD-105, Inspection Level I, Acceptance Quality Level (AQL) 4.0 percent defective.

4.4 Inspection. -

4.4.1 General procedure. - General inspection and test procedures for Government inspection and for contractor inspection shall be in accordance with the provisions of Publication OP 400 and this specification. Lot acceptance inspection shall be conducted at the contractor's plant. Inspections shall be performed at the various stages of assembly best suited for performing the examinations and tests specified herein.

4.4.2 Visual and dimensional examination. - Each rangefinder as specified in 4.3.1.3 shall be visually examined for completeness of manufacture, freedom from defects, proper identification, and workmanship. Dimensional and other nondestructive inspections shall be made in accordance with the acceptance inspection instructions provided by the procuring activity.

4.4.3 Parts. - Parts selected in accordance with 4.3.2, including spare parts, shall be inspected in accordance with the procedures of 4.4.2.

4.4.4 Preservation, packaging, packing, and marking. -

4.4.4.1 Rangefinders. - Each rangefinder container shall be examined to determine that preservation, packing, and marking are in accordance with the requirements of section 5.

4.4.4.2 Parts. - Sample containers of spare parts selected in accordance with 4.3.3 shall be examined to determine that preservation, packaging, packing, and marking are in accordance with the requirements of Section 5.

4.5 Tests and test procedures. -

4.5.1 Preproduction tests. - Preproduction rangefinders (see 4.3.1.1) shall be subjected to the environmental tests of 4.5.3.1 through 4.5.3.4.2 and the lot acceptance tests of 4.5.4. Rangefinders that pass the required tests and meet the requirements specified herein shall be included in the quantity called for in the contract or order.

4.5.2 Periodic tests. - Periodic tests (see 4.3.1.2) shall consist of the environmental tests of 4.5.3.1 through 4.5.3.4.2 and shall be performed on rangefinders which have previously passed the lot acceptance tests of 4.5.4.

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4.5.2.1 Failures during periodic tests. - In the event of failure of any rangefinder during periodic testing, inspection and acceptance of subsequent rangefinders shall be suspended until the reasons for failure and consequent rejection have been determined and the cause of failure corrected by the contractor to the satisfaction of the procuring activity. The contractor shall furnish the procuring activity with a written statement properly endorsed by the Government inspector indicating the reasons for failure and the corrective action taken.

4.5.3 Environmental tests. -

4.5.3.1 Shock stability. - The rangefinder shall be secured to a suitable shock test stand and tested for shock stability to determine conformance to 3.6.1. The blows shall be struck, two in each of the three mutually perpendicular directions.

4.5.3.2 Vibration stability. - The rangefinder shall be tested for vibration stability to determine conformance to 3.6.2. The vibrations shall be applied separately in each of the three mutually perpendicular directions. Each test test in each direction shall be of at least 15 minutes duration.

4.5.3.3 Watertightness. - The rangefinder shall be tested for watertightness to determine conformance to 3.6.3. All exposed surfaces and joints of the change of range and eyepiece assemblies shall be subjected to the floodtight hose test specified in Standard MIL-STD-108. Immediately following the test examination shall be made for the entrance of water into the parts, disassembly being made as necessary.

4.5.3.4 Temperature stability. -

4.5.3.4.1 Temperature extremes. - The rangefinder shall be tested at extremes of temperature to determine conformance to 3.6.4.1. To test at the cold temperature extreme, the rangefinder shall be placed in a cold chamber and allowed to stabilize at minus 20°F at which temperature all rangefinder mechanisms shall be operated. To test at the hot temperature extreme, the rangefinder shall be placed in a hot chamber and allowed to stabilize at plus 120°F at which temperature all rangefinder mechanisms shall be operated. The mechanisms shall be operated in these tests with the rangefinder at 20 degrees depression, 0 degrees elevation, and 95 degrees elevation.

4.5.3.4.2 Temperature variation. - The rangefinder shall be tested for stability of ranging at varying temperature to determine conformance to 3.6.4.2. The rangefinder shall be heated to 30°F above or cooled to 30°F below the temperature of the area in which the ranging measurements are to be taken. The temperature of the rangefinder shall then be allowed to vary and reach the temperature of the test area, the test being started immediately after the change in temperature has begun. This test shall be conducted at high magnification when the freedom from "boiling" is as satisfactory as may

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be expected under the conditions of the test and when the observing conditions are otherwise good. The rangetaker shall make five internal adjustment settings, set the adjuster scale to the mean of the five settings and immediately thereafter take ten readings on a target in the center of the field and at an accurately known range. This procedure, namely five adjuster settings and ten range readings, shall be repeated at least once every twenty minutes during a three hour period or during the period of time required for the rangefinder temperature to stabilize at the temperature of the test area. Five settings shall be made and ten readings taken as indicated above when the temperature of the rangefinder is stabilized at the temperature of the test area. No adjuster settings or range readings shall become known to the rangetaker during this test.

4.5.3.5 Examination and retests following environmental tests. - Following the environmental tests of 4.5.3.1 through 4.5.3.4.2 the rangefinder shall be examined for breakage, loose parts, and other defects. Rangefinders that pass the environmental tests shall be returned to the contractor for any needed repairs and reconditioning. The lot acceptance tests of 4.5.4 shall be repeated as necessary to determine that no changes in mechanical and optical performance have taken place as a result of the environmental tests.

4.5.4 Lot acceptance tests. -

4.5.4.1 Test methods. - Acceptance tests on each assembled rangefinder shall be conducted in accordance with the procedures of Publication NAVORD OD 13521. The test results shall be recorded in the spaces provided in the above ordnance data.

4.5.4.2 Tests. - The acceptance tests shall be performed to determine conformance of each rangefinder to the following requirements of this specification:

3.3.1 Cleanness.
3.3.5 Reticles.
3.3.7 Stray light.
3.3.8 Artificial illumination.
3.3.9 Rotational balance.
3.3.10 Gastightness.
3.4.1 Operation of mechanisms.
3.4.2 Stability of settings.
3.4.3 Change of range mechanism.
3.4.4 Range knob torque.
3.4.5 Orientation of filters.
3.5.1 Image quality.
3.5.2 Resolution.
3.5.3 Alignment of line of sight.
3.5.4 Parallelism of emergent rays.
3.5.5 Tilt of fields.
3.5.6 Equality of magnification.

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3.5.7 Coincidence of focal planes of outside object and reticle markings (freedom from parallax).
3.5.8 Interpupillary adjustment.
3.5.9 Accuracy of diopter graduation.
3.5.10 Correction scale eyepiece focus.
3.5.11 Alignment of inner and outer range scales.
3.5.12 Reproducibility of internal adjuster settings.
3.5.13 Reproducibility of range settings.
3.5.14 Ranging accuracy.
3.5.15 Stability of ranging using various reticle marks.
3.5.16 Dielectric strength.
3.5.17 Insulation resistance.

4.5.4.3 Resubmission. - Rangefinders that fail to pass any lot acceptance test shall be rejected. Rejected rangefinders may be resubmitted for retests in accordance with the procedures of Publication OP 400.

5. PREPARATION FOR DELIVERY

5.1 Preservation. -

5.1.1 Level A. -

5.1.1.1 Cleaning. - The main outer bearings of each rangefinder shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - After cleaning the main outer bearings, any drying required shall be in accordance with procedure D-1 or D-4 of Specification MIL-P-116.

5.1.1.3 Preservation procedure. - After cleaning and drying, the main outer bearings shall be coated with bearing grease conforming to Specification MIL-G-16908.

5.1.2 Level B. - Not applicable.

5.1.3 Level C. - Not applicable.

5.2 Packing. -

5.2.1 Level A. - Each rangefinder shall be packed for stowage or shipment in the stowage box conforming to Drawing LD 413777.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - Not applicable.

5.3 Spare parts. - Unless otherwise specified in the contract or order (see 6.2), spare parts shall be preserved, packaged, and packed in accordance with the plan prepared in conjunction with the NAVORD list and furnished with the contract.

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5.4 Marking. - In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.5 Special Marking. - The preproduction sample (see 4.3.1.1) shall be marked with the following information:

Title, number and date of this specification
Mark and Mod and LD number
Sample for preproduction inspection
Name of manufacturer
Manufacturer's plant address
Reference (contract number)

6. NOTES

6.1 Intended use. - The Rangefinder Mark 75 Mod 1 is intended for use in the Gun Director Mark 68 and Mods as secondary ranging equipment in conjunction with shipboard radar.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Whether preproduction samples and inspection are required (see 3.2 and 4.3.1.1.1)
(c) Testing activity for preproduction samples or samples for period tests if the activity is not designated by the Bureau of Ordnance (see 4.3.1.1 or 4.3.1.2).
(d) Preservation, packaging, and packing of spare parts if different from 5.3.
(e) Special marking when required (see 5.4).

6.3 Unit of error. - The "unit of error" used herein is defined as the error in range, at any given range, which is equivalent to an error of twelve seconds of arc, divided by the magnification, in the angle subtended by the base length of the rangefinder at the given range. Examples of the valuc of one unit of error at various ranges are listed in Table III. The values of unit of error in the table are for a magnification of 24 diameters. For a magnification of 12 diameters the value of the unit of error is twice that shown in the table.

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Table III. - Units of Error

Range (yards)	Unit of error (yards)	Range (yards)	Unit of error (yards)
1000	.66	14000	129.85
1500	1.49	16000	169.60
2000	2.65	18000	214.65
2500	4.14	20000	265.00
3000	5.96	25000	414.00
4000	10.60	30000	596.25
5000	16.56	35000	811.56
6000	23.85	40000	1060.00
8000	42.40	45000	1341.56
10000	66.25	50000	1656.25
12000	95.40		

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